Polynomials — problem 10.65

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A polynomial $f(x) = x^4 + *x^3 + *x^2 + *x + 1$ has three undetermined coefficients denoted by stars. The players A and B move alternately, replacing a star by a real number until all stars are replaced. A wins if all zeros of the polynomial are complex. B wins if at least one zero is real. Show that B can win in spite of his only second move.